Zachary Deutsch

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Education

DUKE UNIVERSITY — Durham, NC
B.S. Mathematics; B.S.E. Mechanical Engineering; Minor in Machine Learning; Minor in Philosophy; GPA: 3.7 — August 2024 – May 2027
Coursework: Calculus I–III; Linear Algebra; Differential Equations; Probability; Statistics & Data Analysis; Data Structures & Algorithms

UNIVERSITY OF OXFORD — Oxford, England
Study Abroad: AI and Ethics — June – August 2025

Experience

DUKE UNIVERSITY, Oca Lab — Durham, NC
Machine Learning Researcher — October 2025 – Present
- Designed experiments to quantify LLM hallucination rates in engineering problem-solving, analyzing error types and detection accuracy on quantitative tasks

DUKE UNIVERSITY, Hickey Lab — Durham, NC
Machine Learning Engineer — November 2024 – Present
- Built and evaluated unsupervised clustering pipelines for multi-organ cell-type annotation using spatial-omics datasets comprising ~220k observations with ~50-dimensional feature vectors

TEL AVIV UNIVERSITY, Data Modeling & Analysis Lab — Tel Aviv, Israel
Machine Learning Research Intern — May – August 2025
- Evaluated quantile-based time-series feature extraction for sEMG classification, comparing accuracy–latency tradeoffs across classifiers and feature representations

SHIRLEY RYAN ABILITYLAB — Chicago, IL
Data Science Intern — June – August 2024
- Applied statistical modeling to outcomes from a $1.5M DoD-funded study to estimate treatment effects

Publications

- **Deutsch, Z.** (2025). *Superintelligence, Instrumental Convergence, and the Limits of AI Apocalypse. AI and Ethics* (Springer Nature).

Conferences

- **Deutsch, Z.**, & Oca, S. R. (2026). *Evaluating engineering students' detection of hallucinations in large language model solutions*. American Society for Engineering Education (ASEE) Annual Conference.
- **Deutsch, Z.**, Miao, Y., & Hickey, J. W. (2025). *Comparative Evaluation of Unsupervised Clustering Methods for Cell-Type Annotation in HuBMAP CODEX Intestinal Tissue* [Poster presentation]. *Duke Health Data Science Conference.*
- **Deutsch, Z.**, Tascesme, E., & Rabin, N. (2025). *sEMG hand and wrist movement time-series classification: Evaluating QUANT and cosine features. Biomedical Engineering Society (BMES) Annual Meeting.*

Skills, Awards, & Interests

Technical: Python (NumPy, pandas); SQL; Machine Learning; Statistics; Java; Excel; PowerPoint
Languages: Japanese (Limited Working Proficiency); Spanish (Limited Working Proficiency)
Awards: 1st Place J.P. Morgan GenerationTech, Google Scholarship, National Youth Science Academy
Interests: Espresso, Liszt, Stoicism, Nietzsche, Cycling